

Acquisition Investment Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Acquisition Investment Discipline (AID), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on AID achieves an annualized gross (net) Sharpe ratio of 0.43 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (16) bps/month with a t-statistic of 2.50 (1.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing) is 22 bps/month with a t-statistic of 2.38.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in returns. This gradual diffusion manifests particularly in corporate investment decisions, where the implications of strategic choices may take time for investors to fully appreciate and incorporate into prices.

We examine how acquisition investment discipline—measured through changes in acquisition-related capital expenditures—predicts future stock returns. Despite acquisitions representing major capital allocation decisions that fundamentally alter firm trajectories, the market appears to underreact to the information contained in acquisition spending patterns. This underreaction creates a persistent return predictability that challenges traditional notions of market efficiency and suggests that even highly visible corporate actions may experience delayed price discovery.

The slow diffusion of information provides a compelling framework for understanding why acquisition investment discipline predicts returns. When firms alter their acquisition spending, they signal important changes in growth strategies, capital allocation efficiency, and management quality. However, limited investor attention constrains the speed at which this information gets incorporated into prices [Hirshleifer and Teoh \(2003\)](#); [DellaVigna and Pollet \(2009\)](#). Investors face cognitive constraints and must allocate scarce attention across numerous information sources, leading to systematic underreaction to value-relevant signals [Peng and Xiong \(2006\)](#).

The complexity of evaluating acquisition decisions exacerbates this slow information processing. Unlike simple earnings announcements, assessing acquisition discipline requires analyzing strategic fit, integration challenges, and long-term value

creation potential. This complexity increases processing costs and delays full incorporation, consistent with models where information acquisition is costly [Grossman and Stiglitz \(1980\)](#); [Verrecchia \(1982\)](#). Moreover, acquisition information often arrives through multiple channels—press releases, regulatory filings, and conference calls—creating dispersion in information arrival that further slows aggregate price discovery [Hong et al. \(2000\)](#).

We predict that return continuation from acquisition investment discipline will concentrate where information frictions are most severe. Small firms with low analyst coverage should exhibit stronger return predictability, as these stocks receive less investor attention and face greater limits to arbitrage [Hong et al. \(2000\)](#); [Zhang \(2006\)](#). Additionally, the gradual diffusion framework predicts that the signal’s predictive power should persist even after controlling for other investment-based anomalies, as each signal may capture distinct aspects of firms’ investment decisions that diffuse through prices at different rates [Cohen et al. \(2003\)](#).

Our empirical analysis reveals that acquisition investment discipline (AID) generates economically and statistically significant return predictability. The value-weighted long/short AID strategy achieves an annualized gross Sharpe ratio of 0.43 and delivers monthly average abnormal returns of 23 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.50. This performance places the strategy in the top 17% of documented anomalies in terms of gross Sharpe ratio, demonstrating that acquisition discipline represents a distinct and powerful predictor of cross-sectional returns.

The predictive power of AID exhibits remarkable robustness across different portfolio construction methodologies and market segments. Among the largest stocks—those above the 80th NYSE percentile—the AID strategy generates average returns of 30 basis points per month with a t-statistic of 2.71, contradicting the common pattern of anomalies concentrating in small, illiquid stocks. The strategy maintains

statistical significance across alternative weighting schemes and breakpoint choices, with thirteen out of twenty-five construction methods yielding t-statistics exceeding two. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy achieves a net Sharpe ratio of 0.35, outperforming 93% of anomalies in the factor zoo.

Controlling for closely related investment-based predictors barely attenuates the AID signal’s explanatory power. When we simultaneously control for six related anomalies—including asset growth, changes in net financial assets, and net external financing—plus the Fama-French six factors, the AID strategy maintains an alpha of 22 basis points per month with a t-statistic of 2.38. This persistence indicates that acquisition discipline captures unique information about firm value that other investment signals fail to subsume. Furthermore, incorporating AID into a composite strategy combining 155 existing anomalies increases terminal wealth by 7.8%, suggesting meaningful out-of-sample benefits for practical portfolio construction.

This paper makes several contributions to the asset pricing literature on investment-based anomalies and information diffusion. While [Cooper et al. \(2008\)](#) document that asset growth predicts returns and [Richardson et al. \(2005\)](#) show that various investment measures contain predictive power, we demonstrate that acquisition-specific investment discipline provides incremental explanatory power beyond these broad measures. Our findings extend [Titman et al. \(2004\)](#) by showing that acquisition discipline, rather than general capital expenditures, drives much of the investment-return relation. Unlike [Lyandres et al. \(2008\)](#), who focus on the investment-external financing interaction, we isolate the pure acquisition component and show it survives controls for financing activities.

Methodologically, we advance the literature by subjecting acquisition investment discipline to the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#), ensuring our results meet the highest standards of robustness in anomaly documen-

tation. Our analysis spans over 50 years of data and examines performance across 212 competing anomalies, providing unprecedented context for evaluating the signal’s economic importance. By demonstrating that AID improves the investment opportunity set even after accounting for transaction costs and competing factors, we address recent concerns about the replicability and implementability of asset pricing anomalies raised in [McLean and Pontiff \(2016\)](#) and [Hou et al. \(2020\)](#).

Our results carry important implications for understanding market efficiency and the limits of arbitrage. The persistence of acquisition investment discipline’s predictive power, especially among large, liquid stocks, challenges pure risk-based explanations and supports behavioral models featuring gradual information diffusion. The signal’s robustness to transaction costs and its ability to enhance multi-factor portfolios suggest that acquisition discipline represents a genuine inefficiency that sophisticated investors can exploit. These findings contribute to the broader debate about market efficiency and highlight the continued importance of fundamental analysis in identifying mispriced securities, even in seemingly efficient segments of the market.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the relevant variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to capture year-over-year changes in acquisition activity and investment patterns. Acquisition Investment Discipline signal relies on two key COMPUSTAT variables. Acquisitions (AQC) represents the cash outflow for acquisitions during the fiscal year, capturing the total amount spent by a firm to acquire other compa-

nies or business units, net of cash acquired. This item reflects the firm’s external growth strategy through mergers and acquisitions activity. Invest advances other (IVAO) represents investments in and advances to unconsolidated subsidiaries, joint ventures, and other investees, including loans and advances to affiliated companies and other long-term investments. This variable captures a broader set of investment activities beyond direct acquisitions. We construct the Acquisition Investment Discipline signal by computing the year-over-year change in acquisitions scaled by lagged investments and advances. Specifically, we calculate $(AQC(t) - AQC(t-1)) / IVAO(t-1)$, where t denotes the fiscal year. This ratio captures the change in a firm’s acquisition spending relative to its existing investment base, providing a measure of how aggressively the firm is expanding its acquisition activity compared to its established investment portfolio. The signal reflects acquisition discipline, with lower values suggesting more restrained acquisition growth relative to the firm’s investment capacity. We use annual observations based on fiscal year-end values and require non-missing values for both AQC and IVAO to ensure accurate calculation of the year-over-year changes. To address potential outliers and ensure economic meaningfulness, we exclude observations where IVAO equals zero or is negative.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the AID signal. Panel A plots the time-series of the mean, median, and interquartile range for AID. On average, the cross-sectional mean (median) AID is -1.81 (0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input AID data. The signal’s interquartile range spans -0.10 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the AID signal for the CRSP universe. On average, the AID signal is available for 2.49% of CRSP names, which on average make up 3.88% of total market

capitalization.

4 Does AID predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on AID using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high AID portfolio and sells the low AID portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short AID strategy earns an average return of 0.27% per month with a t-statistic of 3.05. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.22% to 0.27% per month and have t-statistics exceeding 2.37 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.12, with a t-statistic of 1.90 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 223 stocks and an average market capitalization of at least \$808 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight

small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.02. Out of the twenty-five alphas reported in Panel A, the t-statistics for thirteen exceed two, and for two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-31bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.17. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the AID trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the AID strategy perfor-

mance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and AID, as well as average returns and alphas for long/short trading AID strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the AID strategy achieves an average return of 30 bps/month with a t-statistic of 2.71. Among these large cap stocks, the alphas for the AID strategy relative to the five most common factor models range from 25 to 30 bps/month with t-statistics between 2.20 and 2.65.

5 How does AID perform relative to the zoo?

Figure 2 puts the performance of AID in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the AID strategy falls in the distribution. The AID strategy’s gross (net) Sharpe ratio of 0.43 (0.35) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the AID strategy (red line).² Ignoring trading costs, a \$1 invested in the AID strategy would have yielded \$3.16 which ranks the AID strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the AID strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

would have yielded \$2.16 which ranks the AID strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the AID relative to those. Panel A shows that the AID strategy gross alphas fall between the 52 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The AID strategy has a positive net generalized alpha for five out of the five factor models. In these cases AID ranks between the 70 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does AID add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of AID with 203 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price AID or at least to weaken the power AID has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of AID conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AID} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AID}AID_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AID,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on AID. Stocks are finally grouped into five AID portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AID trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on AID and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the AID signal in these Fama-MacBeth regressions exceed -1.88, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on AID is -1.82.

Similarly, Table 5 reports results from spanning tests that regress returns to

an average month.

the AID strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the AID strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.18, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the AID trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.38.

7 Does AID add relative to the whole zoo?

Finally, we can ask how much adding AID to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the AID signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes AID grows to \$1071.39.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which AID is available.

8 Conclusion

This study provides compelling evidence that Acquisition Investment Discipline (AID) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that AID-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on AID delivers an annualized gross Sharpe ratio of 0.43, which remains economically meaningful at 0.35 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 23 basis points gross (16 basis points net) relative to the Fama-French five-factor model plus momentum, with statistically significant t-statistics of 2.50 and 1.80 respectively, underscores its practical relevance for portfolio management. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of twelve factors, including six closely related investment-based strategies from the factor zoo. The persistence of a 22 basis point monthly alpha with a t-statistic of 2.38 in this stringent specification suggests that AID captures unique information about firm value creation through acquisition activities that is not subsumed by existing factors.

These results have important implications for both academic research and investment practice. For academics, our findings contribute to the growing literature on investment-based anomalies and suggest that the discipline exercised in acquisition decisions contains distinct information about future returns beyond traditional measures of corporate investment. For practitioners, the net-of-cost performance metrics indicate that AID-based strategies remain viable even after accounting for implementation frictions, though the reduction from gross to net Sharpe ratios highlights the importance of efficient execution.

Several limitations of this study warrant consideration and point to avenues for future research. First, while our analysis focuses on U.S. equity markets, the generalizability of AID’s predictive power to international markets remains an open question. Future research should examine whether similar patterns exist in other developed and emerging markets, accounting for differences in MA regulations and market structures. Second, our study period may not fully capture the signal’s performance across different market regimes, particularly in environments characterized by changing interest rates or varying levels of MA activity. Third, while we control for numerous factors, the economic mechanisms driving the AID premium deserve deeper investigation. Future studies could explore whether the signal’s predictive power stems from risk-based explanations, behavioral biases in how investors process acquisition information, or limits to arbitrage that prevent rapid price correction.

Additionally, future research could investigate the interaction between AID and corporate governance characteristics, examine the signal’s performance conditional on deal characteristics such as payment method or target status, and explore potential enhancements through machine learning techniques or alternative signal constructions. Understanding the time-varying nature of the AID premium and its relationship with macroeconomic conditions could also provide valuable insights for dynamic portfolio allocation strategies.

In conclusion, this paper establishes Acquisition Investment Discipline as a robust and economically significant predictor of equity returns that merits inclusion in the canon of documented asset pricing anomalies. While implementation costs reduce but do not eliminate its profitability, the signal’s ability to generate alpha relative to a comprehensive factor model suggests it captures unique information valuable to both researchers and practitioners in understanding the cross-section of expected returns.

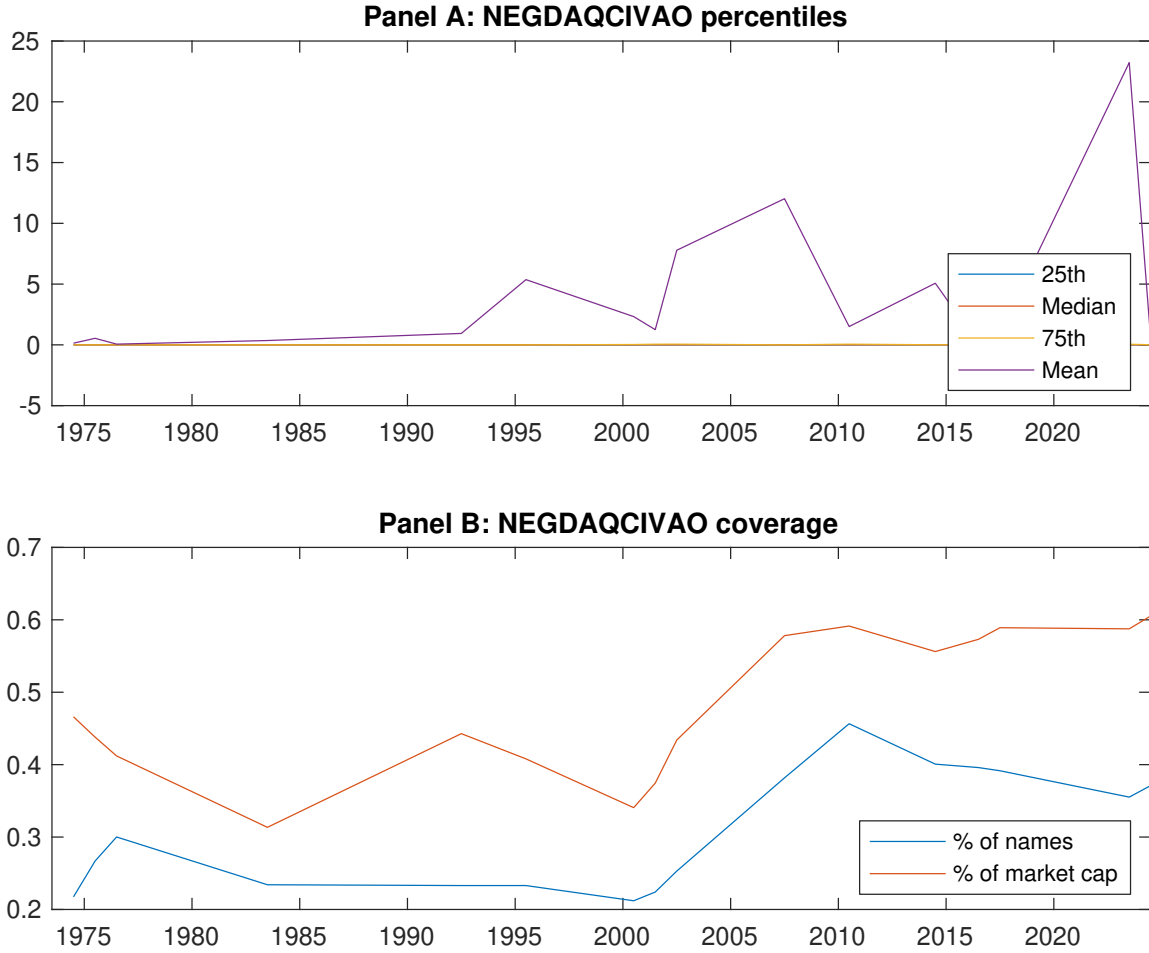


Figure 1: Times series of AID percentiles and coverage. This figure plots descriptive statistics for AID. Panel A shows cross-sectional percentiles of AID over the sample. Panel B plots the monthly coverage of AID relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on AID. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on AID-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.84]	0.80 [4.14]	0.56 [3.06]	0.59 [2.97]	0.82 [4.18]	0.27 [3.05]
α_{CAPM}	-0.10 [-1.53]	0.17 [2.29]	-0.02 [-0.27]	-0.06 [-0.85]	0.18 [2.45]	0.27 [3.08]
α_{FF3}	-0.10 [-1.59]	0.11 [1.52]	-0.09 [-1.12]	-0.10 [-1.44]	0.16 [2.17]	0.26 [2.86]
α_{FF4}	-0.10 [-1.62]	0.12 [1.68]	-0.13 [-1.71]	-0.09 [-1.29]	0.17 [2.25]	0.27 [2.94]
α_{FF5}	-0.18 [-2.85]	0.06 [0.85]	-0.11 [-1.33]	-0.15 [-2.07]	0.04 [0.53]	0.22 [2.37]
α_{FF6}	-0.18 [-2.77]	0.08 [1.04]	-0.14 [-1.78]	-0.14 [-1.86]	0.06 [0.76]	0.23 [2.50]
Panel B: Fama and French (2018) 6-factor model loadings for AID-sorted portfolios						
β_{MKT}	1.02 [68.93]	1.02 [60.45]	0.93 [50.13]	1.05 [62.07]	1.04 [61.64]	0.01 [0.51]
β_{SMB}	-0.00 [-0.19]	-0.10 [-4.01]	0.02 [0.59]	-0.09 [-3.68]	-0.01 [-0.39]	-0.01 [-0.17]
β_{HML}	-0.04 [-1.27]	0.17 [5.28]	0.18 [5.05]	0.05 [1.43]	-0.04 [-1.35]	-0.01 [-0.18]
β_{RMW}	0.15 [5.26]	0.07 [2.02]	0.00 [0.00]	0.00 [0.11]	0.19 [5.74]	0.04 [0.84]
β_{CMA}	0.11 [2.45]	0.09 [1.83]	0.04 [0.79]	0.20 [4.11]	0.23 [4.62]	0.12 [1.90]
β_{UMD}	-0.01 [-0.34]	-0.02 [-1.34]	0.06 [3.10]	-0.02 [-1.31]	-0.03 [-1.63]	-0.02 [-1.03]
Panel C: Average number of firms (n) and market capitalization (me)						
n	223	292	329	292	223	
me (\$10 ⁶)	1314	1636	808	1455	1384	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the AID strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.05]	0.27 [3.08]	0.26 [2.86]	0.27 [2.94]	0.22 [2.37]	0.23 [2.50]
Quintile	NYSE	EW	0.13 [2.02]	0.15 [2.34]	0.11 [1.82]	0.08 [1.25]	0.08 [1.20]	0.05 [0.82]
Quintile	Name	VW	0.14 [1.75]	0.16 [1.90]	0.14 [1.64]	0.16 [1.91]	0.09 [1.05]	0.11 [1.34]
Quintile	Cap	VW	0.17 [1.84]	0.17 [1.82]	0.17 [1.86]	0.23 [2.44]	0.14 [1.50]	0.19 [2.01]
Decile	NYSE	VW	0.36 [3.13]	0.37 [3.15]	0.33 [2.85]	0.34 [2.87]	0.28 [2.29]	0.29 [2.36]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.53]	0.22 [2.42]	0.20 [2.20]	0.20 [2.26]	0.16 [1.72]	0.16 [1.80]
Quintile	NYSE	EW	-0.08 [-1.17]					
Quintile	Name	VW	0.10 [1.20]	0.10 [1.26]	0.08 [0.98]	0.10 [1.14]	0.03 [0.41]	0.05 [0.58]
Quintile	Cap	VW	0.12 [1.35]	0.11 [1.22]	0.11 [1.20]	0.14 [1.55]	0.08 [0.85]	0.10 [1.12]
Decile	NYSE	VW	0.31 [2.67]	0.30 [2.54]	0.26 [2.24]	0.27 [2.28]	0.20 [1.69]	0.21 [1.77]

Table 3: Conditional sort on size and AID

This table presents results for conditional double sorts on size and AID. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on AID. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high AID and short stocks with low AID. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	AID Quintiles					AID Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.94 [3.38]	0.93 [3.66]	0.98 [3.67]	0.83 [3.11]	0.74 [2.71]	-0.20 [-1.98]	-0.20 [-1.93]	-0.25 [-2.44]	-0.21 [-1.99]	-0.30 [-2.84]	-0.26 [-2.49]
	(2)	0.79 [2.92]	0.82 [3.21]	0.84 [3.42]	0.91 [3.60]	0.98 [3.72]	0.18 [1.50]	0.22 [1.82]	0.14 [1.19]	0.08 [0.65]	-0.01 [-0.09]	-0.05 [-0.41]
	(3)	0.88 [3.59]	0.78 [3.36]	0.89 [3.96]	1.01 [4.03]	0.88 [3.52]	-0.00 [-0.01]	-0.01 [-0.11]	-0.01 [-0.07]	-0.05 [-0.42]	-0.01 [-0.09]	-0.04 [-0.33]
	(4)	0.83 [3.69]	0.76 [3.44]	0.85 [4.05]	0.74 [3.45]	0.85 [3.78]	0.02 [0.22]	0.02 [0.17]	-0.02 [-0.17]	0.04 [0.38]	-0.04 [-0.40]	0.01 [0.05]
	(5)	0.51 [2.66]	0.74 [3.72]	0.52 [2.86]	0.54 [2.75]	0.81 [4.10]	0.30 [2.71]	0.29 [2.63]	0.30 [2.65]	0.29 [2.58]	0.25 [2.20]	0.25 [2.20]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	AID Quintiles					AID Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	139	138	138	139	139	16	12	12	12	15	
	(2)	42	42	41	42	41	26	26	25	26	26	
	(3)	32	32	32	32	32	49	49	49	49	49	
	(4)	29	29	29	29	29	116	115	111	112	115	
(5)	31	31	31	31	31	938	1425	888	1270	1067		

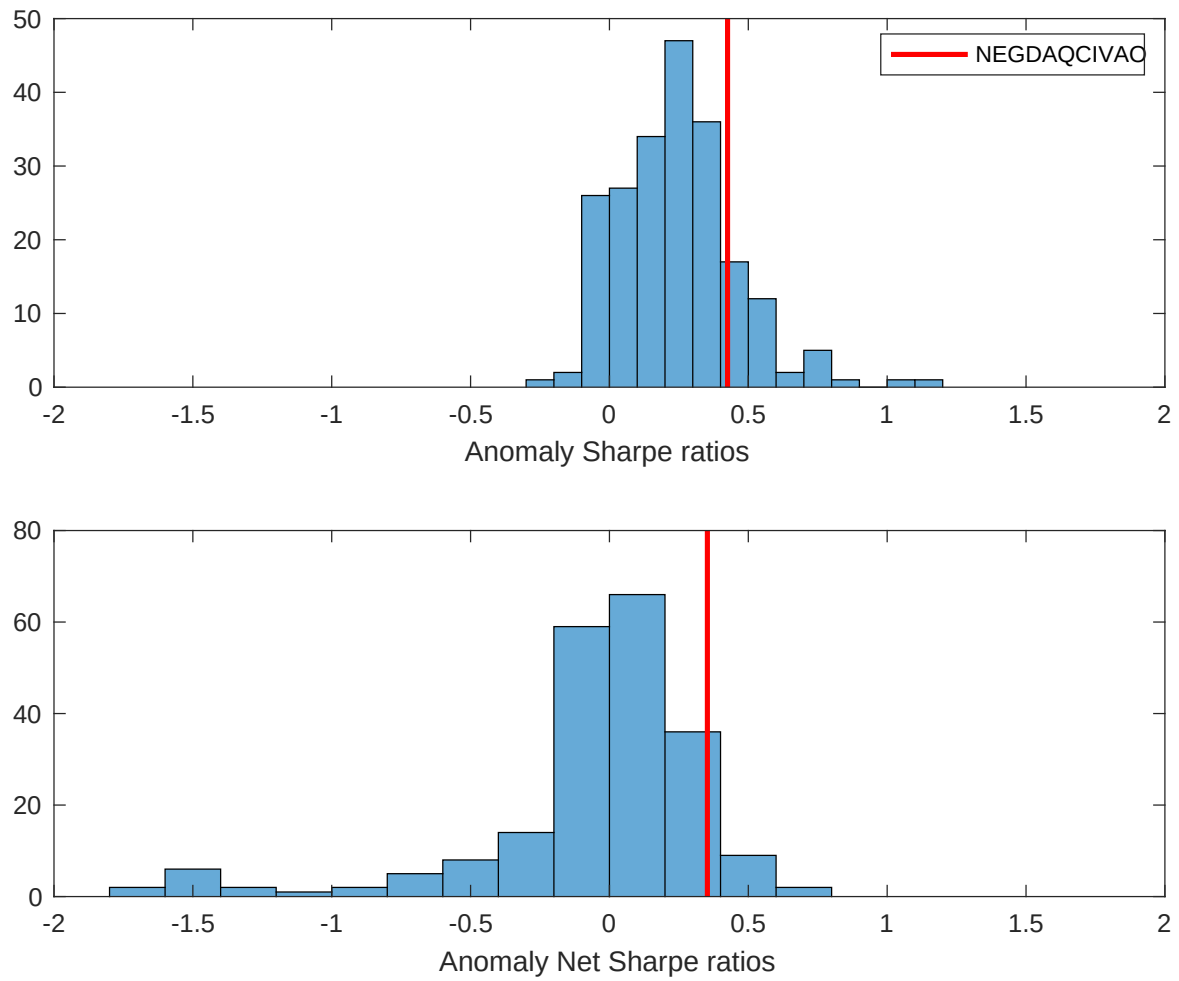


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the AID with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

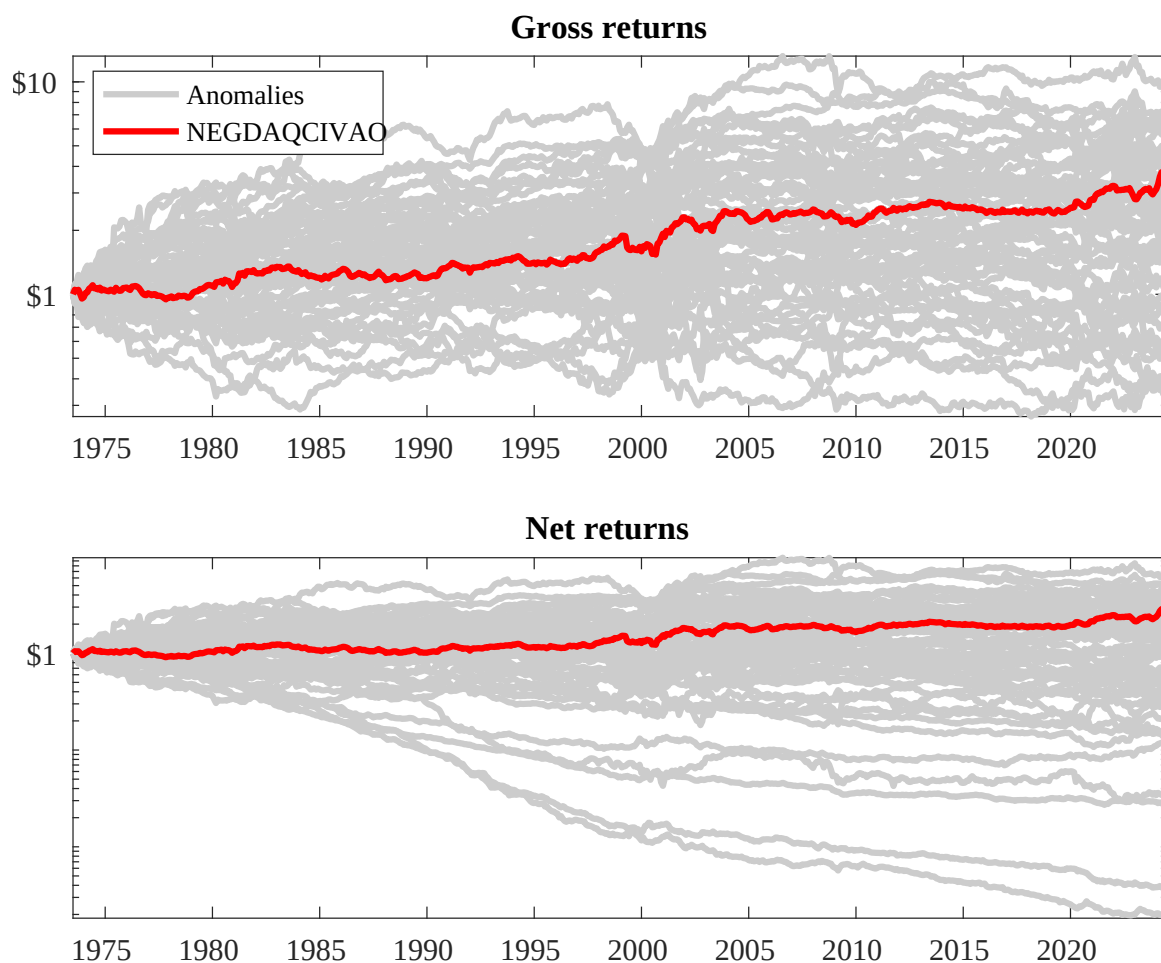
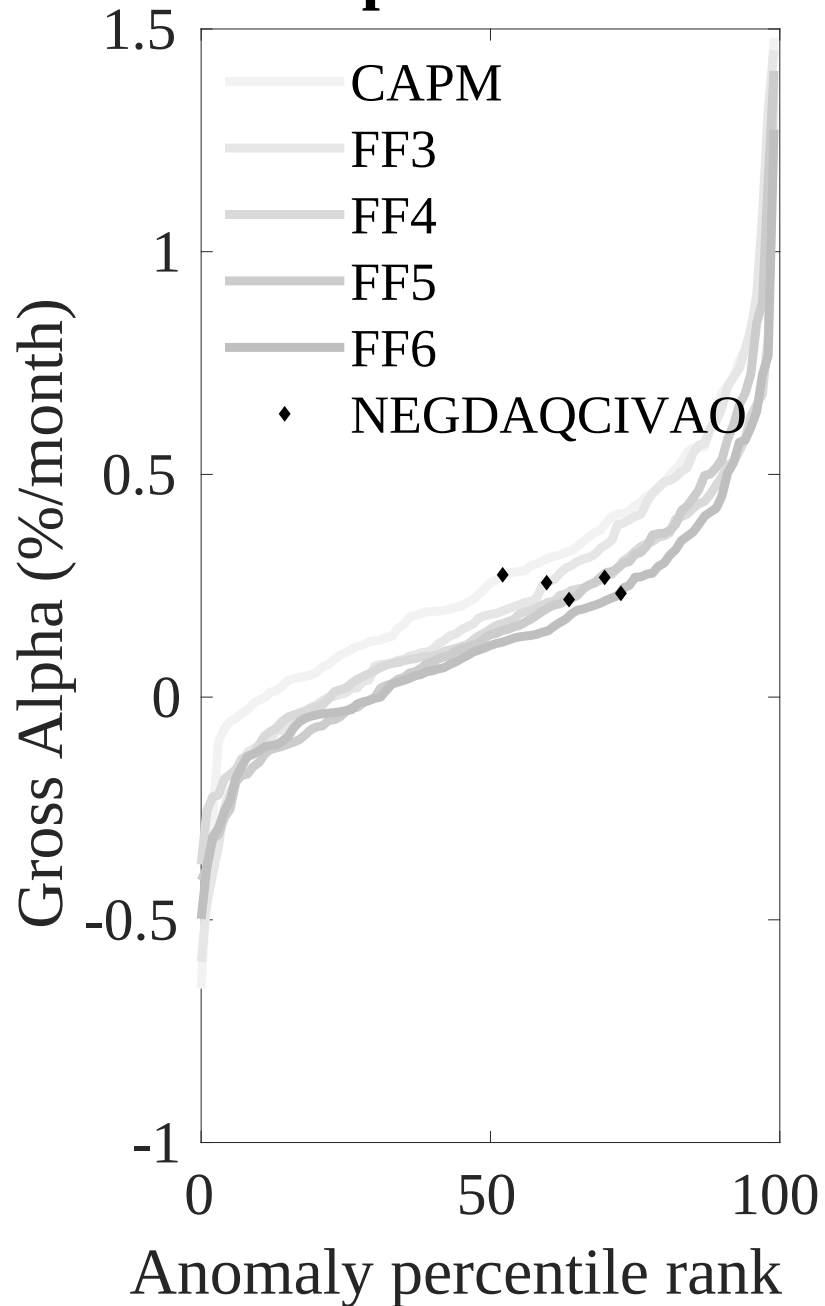


Figure 3: Dollar invested.

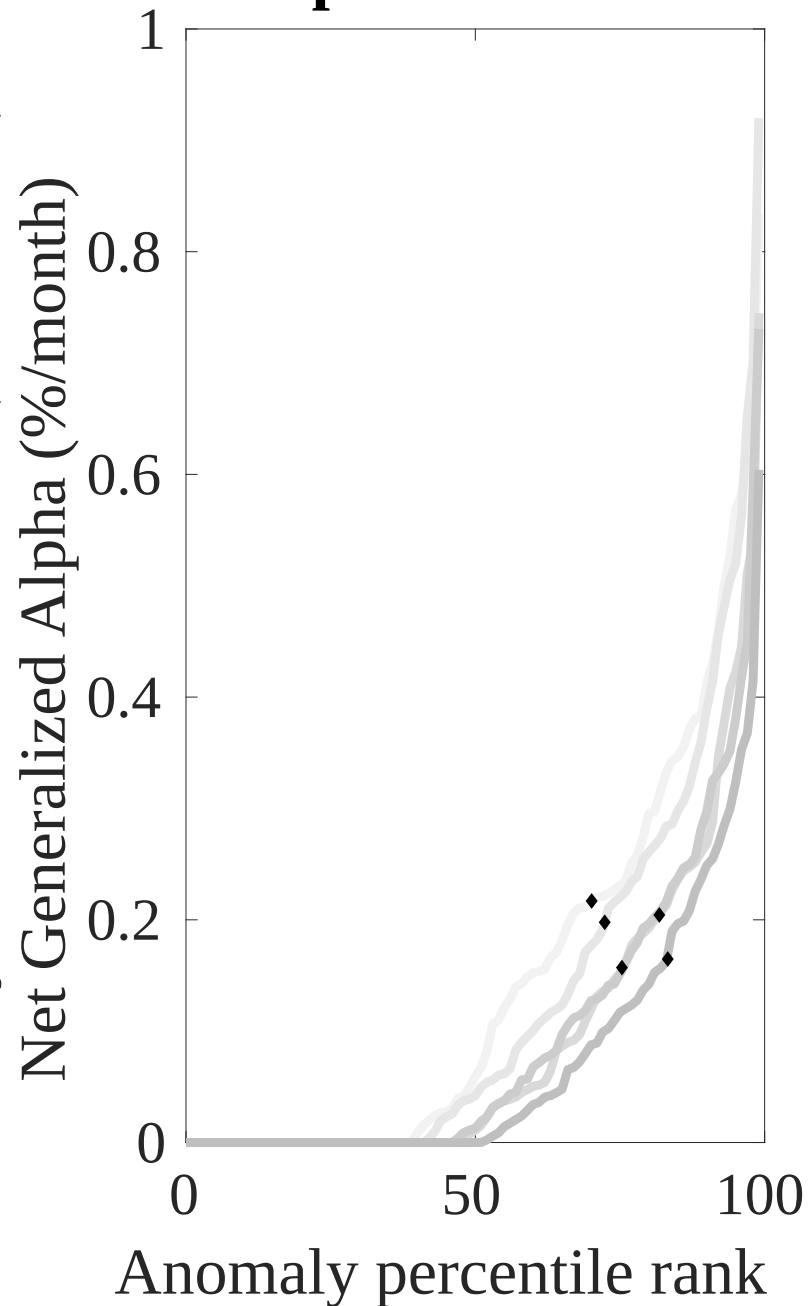
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the AID trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

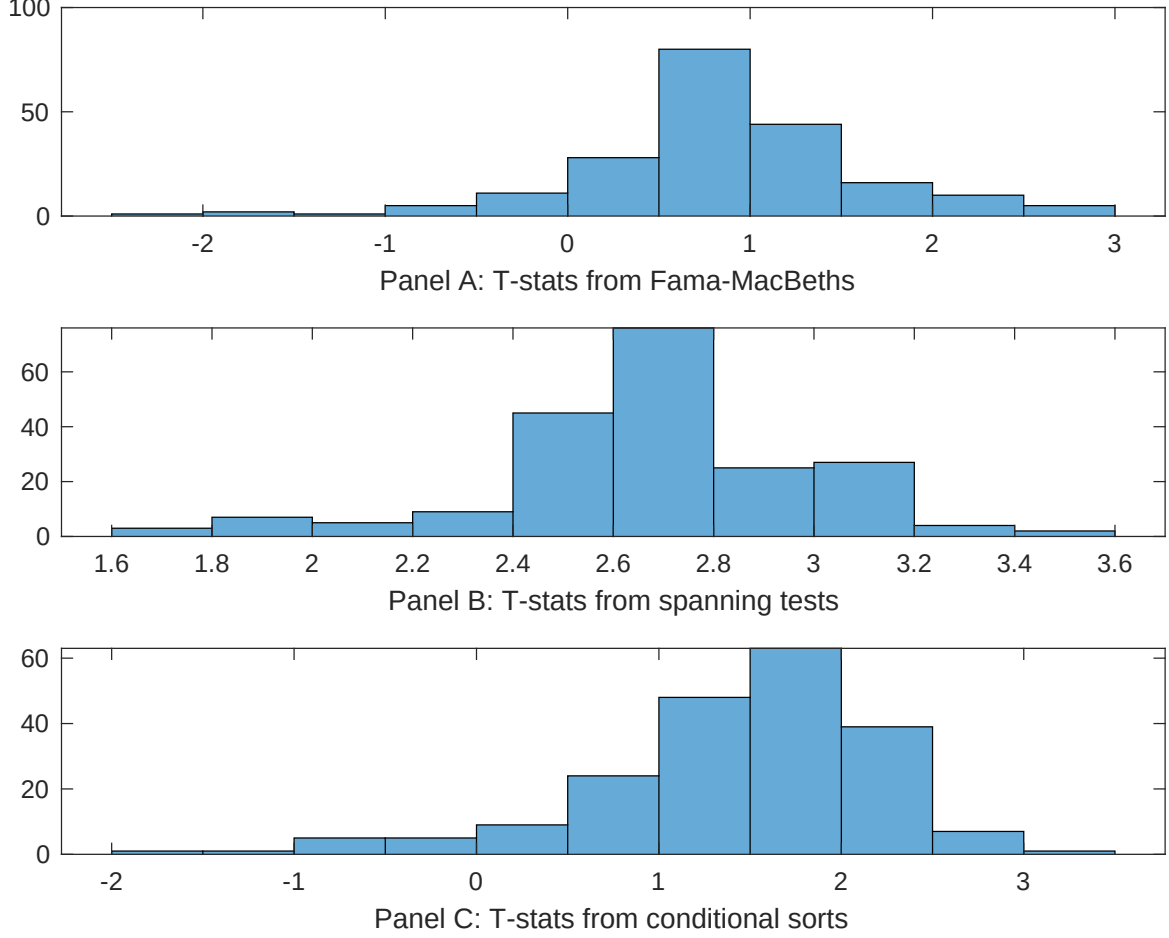


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of AID conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AID} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AID}AID_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AID,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on AID. Stocks are finally grouped into five AID portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AID trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on AID. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{AID}AID_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [6.14]	0.13 [5.60]	0.13 [5.71]	0.18 [7.00]	0.13 [5.66]	0.13 [5.81]	0.12 [5.26]
AID	-0.66 [-1.88]	0.25 [0.07]	0.13 [0.37]	0.84 [0.24]	-0.11 [-0.03]	-0.17 [-0.47]	-0.67 [-1.82]
Anomaly 1	0.11 [7.96]						0.74 [4.65]
Anomaly 2		0.52 [2.52]					0.29 [0.13]
Anomaly 3			0.16 [4.21]				0.73 [1.23]
Anomaly 4				0.48 [4.58]			-0.19 [-1.37]
Anomaly 5					0.17 [4.42]		0.35 [0.95]
Anomaly 6						0.18 [4.99]	0.93 [2.19]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the AID trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{AID} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [2.28]	0.24 [2.58]	0.21 [2.29]	0.21 [2.24]	0.22 [2.35]	0.20 [2.18]	0.22 [2.38]
Anomaly 1	26.10 [4.91]						22.96 [3.53]
Anomaly 2		-12.00 [-2.72]					-7.67 [-1.51]
Anomaly 3			13.75 [3.00]				3.72 [0.51]
Anomaly 4				11.90 [2.38]			-5.42 [-0.73]
Anomaly 5					6.20 [1.36]		-0.74 [-0.15]
Anomaly 6						9.64 [2.07]	5.23 [1.08]
mkt	1.32 [0.63]	1.05 [0.49]	1.19 [0.56]	1.70 [0.79]	1.20 [0.56]	2.62 [1.17]	1.68 [0.76]
smb	-2.24 [-0.70]	-1.02 [-0.32]	-0.29 [-0.09]	-0.70 [-0.22]	1.00 [0.30]	2.76 [0.77]	-0.75 [-0.20]
hml	-2.07 [-0.51]	-0.54 [-0.13]	-2.65 [-0.64]	-2.17 [-0.52]	-3.34 [-0.73]	0.28 [0.07]	-0.75 [-0.17]
rmw	4.22 [1.02]	2.69 [0.64]	5.36 [1.28]	4.52 [1.08]	4.84 [1.14]	-1.63 [-0.32]	0.23 [0.04]
cma	-19.02 [-2.14]	9.40 [1.48]	-1.18 [-0.15]	1.12 [0.14]	9.35 [1.40]	6.22 [0.89]	-18.74 [-2.01]
umd	-1.68 [-0.80]	-1.98 [-0.93]	-2.09 [-0.99]	-2.51 [-1.18]	-1.96 [-0.92]	-2.27 [-1.06]	-1.50 [-0.71]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	4	2	2	1	1	1	4

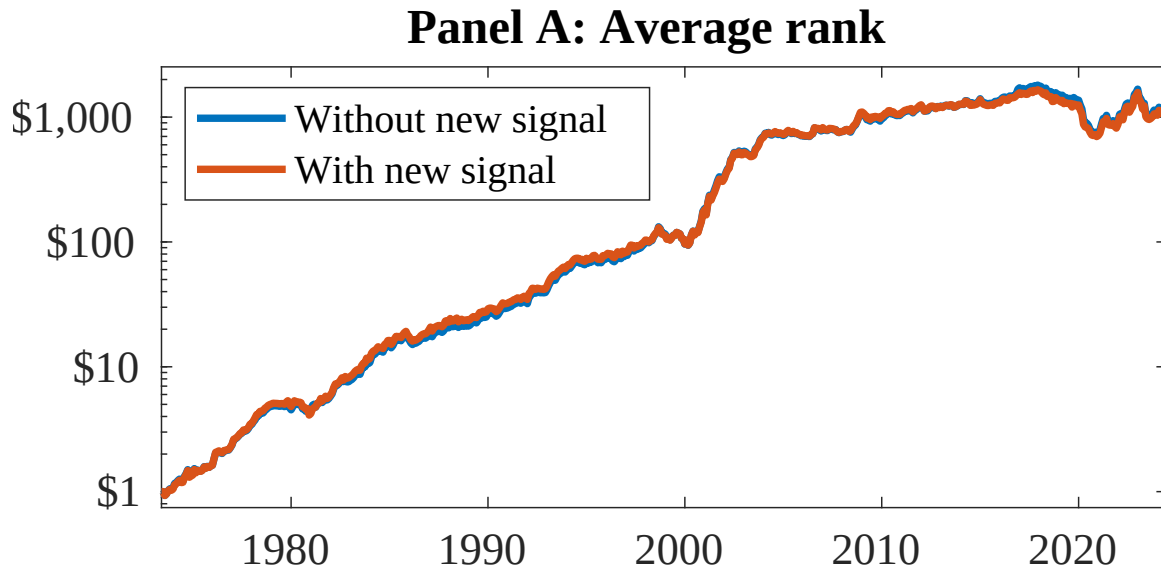


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as AID. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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